



Summary

Pricing options under stochastic volatility

by

Rudolf Gerrit Venter

Submitted in partial fulfillment of the requirements for the degree

Magister Scientiae in Mathematics of Finance

in the Faculty of Natural & Agricultural Sciences

University of Pretoria

Pretoria

January 2003



Summary

In this dissertation some of the real world deviations from the assumptions made in the Black-Scholes option pricing framework is investigated. Special attention is paid to volatility, the standard deviation of stock price returns. Unlike the assumption of constant volatility of increments in Brownian motion, volatility in the market is stochastic. Market models allowing for stochastic volatility are no longer complete as in the Black-Scholes framework. Options in incomplete markets are harder to price since investors demand higher returns for taking additional risk.

Duan (1995) proposed an option pricing measure for incomplete markets, due to stochastic volatility, called the Local Risk-Neutral Valuation Relationship (LRNVR). Under the LRNVR, the local risk neutral measure (Q) is equivalent to the real world measure (P), the conditional expected return under the Q measure equals the risk-free rate and the conditional one period ahead variances under both measures are equal, P almost surely. The LRNVR holds for consumers with familiar utility functions.

Stock returns are assumed to follow a Generalized Autoregressive Conditional Heteroscedastic (GARCH) process. This process is a discrete time statistical time series that is calibrated over stock returns. In this dissertation the LRNVR and related option pricing methodology is comprehensively investigated.

Warrants traded on the JSE Securities Exchange violates the Black-Scholes assumptions in two additional ways, short selling is restricted and the market is somewhat illiquid. One of the results of these violations is that the standard deviation and the implied volatility, volatility implied by the market price of the option, are out of sync. The implied volatility tends to be higher than the volatility of stock market returns.

In this dissertation the GARCH option pricing process is applied to the implied volatility of the warrant instead of the stock price process, as done by Duan. This method compares well with the use of implied volatility to price warrants.



Acknowledgements

I would like to thank my supervisors, Professor Frik van Niekerk of the University of Pretoria and Johan Verwey of Gensec Bank for their support and guidance. My thanks to Professor Dawie de Jongh for advice and references to literature is also due. I am also grateful to my family for their motivation and support.



Contents

Glossary of notation	v
1 Introduction	1
1.1 The Problem of Stochastic Volatility	2
1.2 A Proposed Solution	3
1.3 Description of South African Derivative Instruments and Experiment	4
1.4 Outline of the Dissertation	4
I Background	6
2 Some Probability Essentials	7
2.1 Introduction	7
2.2 Probability Space	7
2.2.1 Probability Space	7
2.2.2 σ -algebra	7
2.2.3 Borel Sets in $\mathbb{R}$	8
2.2.4 Filtration	8
2.2.5 Measurability and Adaptedness	8
2.2.6 Almost everywhere	9
2.3 Moments and Stationarity	9
2.3.1 Expected Value	9
2.3.2 Conditional Expectation	10
2.3.3 Variance, Conditional Variance and Standard Deviation	10
2.3.4 Covariance and Autocovariance	12
2.3.5 Correlation and Autocorrelation	13
2.3.6 Lag	13
2.3.7 Higher Moments	13
2.3.8 Stationarity	13
2.4 Cumulative Distribution Function and Probability Density Function	14
2.4.1 Joint Continuous Distributions	15

CONTENTS

ii

2.5	The Normal Distribution and its Moment Generating Function	16
2.5.1	The Normal Distribution	16
2.5.2	Moments of the Normal Distribution	17
2.5.3	Chi-square Distribution	20
2.6	The Return Series and Lognormal Distribution	22
2.6.1	Returns Series	22
2.6.2	The Arithmetic Returns Series	24
2.6.3	The Geometric Returns Series	24
2.6.4	Lognormal Distribution	24
2.7	Hypothesis Testing	25
2.7.1	Jarque-Bera Test for Normality	25
2.7.2	Autocorrelation	26
2.7.3	Volatility Clustering	27
2.7.4	The Leverage Effect	27
3	An Introduction to Time Series Models	29
3.1	Objectives	29
3.2	Preliminaries	29
3.2.1	White Noise	29
3.2.2	Linear Time Series	29
3.2.3	Lag Operators and Difference Operators	30
3.3	Autoregressive Process (AR)	30
3.4	Moving Averages Process (MA)	31
3.5	Autoregressive Moving Averages (ARMA)	31
3.6	Stationarity of ARMA Processes	32
3.7	Estimation of ARMA Parameters	34
4	Univariate Volatility Processes	36
4.1	Objectives	36
4.2	Exponentially Weighted Moving Averages	36
4.2.1	RiskMetrics	37
4.3	Generalized Conditional Autoregressive Conditional Heteroscedasticity	38
4.4	GARCH(p, q)	38
4.4.1	Stationarity	38
4.4.2	Stylized Facts	40
4.4.3	Estimation of GARCH Regression Model	41
4.5	Integrated GARCH	43
4.6	GARCH-in-Mean	44
4.7	Asymmetric GARCH and the Leverage Effect	45
4.7.1	Exponential GARCH	45
4.7.2	Asymmetric GARCH	45
4.7.3	Glosten, Jagannathan and Runkle GARCH	45
4.8	Limitations of the GARCH Process	45



CONTENTS

iii

II Risk-Neutral Valuation 47

5 Risk-Neutral Valuation	48
5.1 Objectives	48
5.2 Essentials of Continuous-time Stochastic Calculus	49
5.2.1 Brownian Motion	49
5.2.2 Martingales	49
5.2.3 Ito Process	50
5.2.4 Ito Formula (in 1-Dimension)	50
5.2.5 Absolute Continuous	51
5.2.6 Radon-Nikodym	51
5.2.7 Risk-neutral Probability Measure	52
5.2.8 Girsanov's Theorem in One Dimension	52
5.3 Continuous-time Finance Essentials	53
5.3.1 Self-financing	54
5.3.2 Admissible Trading Strategy	55
5.3.3 Attainable Claim	55
5.3.4 Arbitrage Opportunity	55
5.3.5 Complete Market	55
5.4 Risk-Neutral Valuation under Constant Volatility	55
5.4.1 The Stock Price Process	56
5.4.2 The Discounted Stock Price Process	57
5.4.3 Girsanov's Theorem Applied	57
5.4.4 Pricing Options under Constant Volatility	58
5.4.5 The Black-Scholes Formula and Implied Volatility	60

III Option pricing under the Local Risk-Neutral Valuation Relationship 62

6 Local Risk-Neutral Valuation	63
6.1 Introduction	63
6.2 The Stock Price Process in Discrete Time	64
6.3 The Stock Price Model under certain GARCH Volatility	65
6.4 Consumer Utility Essentials	66
6.4.1 Utility Functions	66
6.4.2 Risk Aversion	67
6.5 A General Consumption-Investment Strategy	68
6.6 The Local Risk-Neutral Valuation Relationship	69
6.7 The Local Risk-Neutral Probability Measure	70
6.8 The Stock Price Process under LRNVR	78



CONTENTS

iv

7	GARCH Option Pricing and Hedging	82
7.1	Introduction	82
7.2	Option Pricing under the LRNVR	82
7.3	Some Properties of the GARCH(1, 1) Process under LRNVR	85
IV	Implementation and Numerical Results	95
8	Implementation of GARCH Option Pricing	96
8.1	Introduction	96
8.2	Calibrating the GARCH Process to Empirical Data	96
8.2.1	Historical Data	96
8.2.2	Implied Volatility	97
8.3	Monte Carlo Simulations	98
8.3.1	European Option with Constant Volatility	99
8.3.2	European Options with GARCH Volatility	99
8.3.3	Notes	100
8.3.4	Generating Other Distributions from the Uniform Dis- tribution	101
8.4	Variance Reduction Techniques	101
8.4.1	The Antithetic Variable Technique	102
8.4.2	Moment Matching	102
9	Study and Results	103
9.1	Aim	103
9.2	Methodology and Data	103
9.3	Measures of Results	104
9.4	Results	106
9.4.1	The Results:	106
9.4.2	Conclusion to Results	122
9.4.3	Comments on Study and Results	122
10	Conclusion	123
11	Related Literature	124



Glossary of notation

Glossary of frequently used notation:

$(\Omega, \mathcal{F}, P)$, 7
 $a.e.$, 9
 $A(L)$, 38
 $B(L)$, 38
cdf, 14 (Cumulative distribution function)
 $cor[X, Y]$, 13
 $cov[X, Y]$, 12
 $E(e^{tX})$, 17
 $E[X]$, 9
 $E[X | \Phi]$, 10
 $F(x)$, 14 (Cumulative distribution function)
 $f(x)$, 14 (Probability density function)
 $\chi^2(v)$, 20
 $L^1(\Omega, \mathcal{F}, P)$, 9
 $M_X(t)$, 17
 $N(\mu, \sigma^2)$, 16
pdf, 14 (Probability density function)
 $Std[X]$, 10
 σ_t^2 , 38 (GARCH process)
 $u(x)$, 66
 $Var[X]$, 10
 $Var[X | H]$, 10